Addressing the Yang–Mills Existence and Mass-Gap Problem

Through Relational Rigidity and Phenomenon-Preserving Representations

Abstract

The phenomenon-preserving interpretation of physical representation suggests a concrete reformulation of the Yang–Mills existence and mass-gap problem. Rather than privileging gauge potentials, particles, or a particular regulator, one seeks a coherent continuum family of gauge-invariant observable relations and proves that local non-Abelian dynamics generates a strictly positive form of infrared relational rigidity. This rigidity must be transported through gauge-invariant coarse-graining, the continuum and infinite-volume limits, and Osterwalder–Schrader reconstruction to the spectrum of the physical Hamiltonian. The programme developed here isolates three theorem targets: a rigidity-transfer theorem, an RG-to-rigidity theorem, and a Yang–Mills infrared-basin-entry theorem. The first is a conditional spectral principle; the second can be calibrated in restricted models; the third contains the genuine four-dimensional frontier. No result in this essay constitutes a proof of the Clay problem; the purpose is to identify a noncircular proof architecture and the exact generative theorem still required.

1 The Clay problem under the relational interpretation

The official target is to prove, for every compact simple gauge group G, that a nontrivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a physical Hamiltonian H satisfying

$$\mathrm{Spec}(H) \subseteq \{0\} \cup [\Delta, \infty), \qquad \Delta > 0. \tag{1}$$

The construction must satisfy sufficiently strong axiomatic quantum-field theoretic conditions [1]. The difficulty includes both the existence of a nontrivial four-dimensional continuum theory and a gap uniform in the infinite-volume limit.

Under the relational ontology, the problem becomes:

> Construct a coherent continuum family of gauge-invariant observable relations, and prove that local non-Abelian dynamics generates a strictly positive form of infrared relational rigidity that survives every representation change needed to reconstruct the physical Hamiltonian.

This divides the problem into three logically distinct obligations:

Generation:	local non-Abelian constraints generate infrared rigidity;
Transport:	the rigidity survives changes of scale and representation;
Realization:	the coherent limiting structure reconstructs a physical QFT.

(2)

2 A gauge-invariant pre-Hamiltonian rigidity object

The mass gap is naturally stated in Hamiltonian language, but the Hamiltonian is available only after the continuum theory has been constructed. A proof therefore needs a pre-Hamiltonian manifestation of the same physical rigidity.

Let

$$\mathfrak{R}_{a,L} = \left(\mathcal{A}_{a,L}^{\mathrm{inv}}, \omega_{a,L}, \Theta, \tau, \mathcal{L}\right) \tag{3}$$

denote a regulated Yang–Mills representation. Here

- $\mathcal{A}_{a,L}^{\mathrm{inv}}$ is a gauge-invariant observable algebra;
- $\omega_{a,L}$ is the Wilson-lattice state;
- Θ is Euclidean time reflection;
- τ is Euclidean translation;
- $\mathcal{L}$ records cutoff, volume, and renormalization maps.

For a centered gauge-invariant observable A, define the Euclidean temporal correlation

$$C_{a,L}^{A}(t) = \omega_{a,L}(A^{*}\tau_t(A)), \qquad \omega_{a,L}(A) = 0. \tag{4}$$

Reflection positivity gives this expression a spectral interpretation after Osterwalder–Schrader reconstruction.

Definition 1 (Uniform physical relational rigidity)**.** *A family $\{\mathfrak{R}_{a,L}\}$ has uniform physical relational rigidity with rate $m_* > 0$ if, for every $m < m_*$ and every A in a convergence-determining gauge-invariant class,*

$$\left|C_{a,L}^{A}(t)\right| \le K_A e^{-mt}, \tag{5}$$

where t is measured in physical units, K_A is stable under the required renormalization of A, the estimate is uniform in L, it persists along the continuum trajectory $a \to 0$, and the observable class becomes cyclic or spectrally complete after reconstruction.

This property is formulated without treating A_μ, a particular gauge fixing, perturbative particles, or a finite-lattice Hamiltonian as fundamental. It records the persistence of physical distinctions in a gauge-invariant observable language.

3 A conditional rigidity-transfer theorem

The first exact theorem target is a transport result from regulated Euclidean rigidity to the continuum physical spectrum.

Theorem 1 (Relational Rigidity Transfer; conditional form)**.** *Suppose that:*

(i) regulated gauge-invariant Schwinger functions converge to a nontrivial continuum family;

(ii) the limiting family satisfies the Osterwalder–Schrader axioms;

(iii) uniform physical relational rigidity holds with rate $m_* > 0$;

(iv) the observables satisfying the estimate generate a dense subspace of the reconstructed physical Hilbert space orthogonal to the vacuum.

Then the reconstructed Hamiltonian obeys

$$\mathrm{Spec}(H) \subseteq \{0\} \cup [m_*, \infty). \tag{6}$$

Proof mechanism. Osterwalder–Schrader reconstruction identifies the limiting Euclidean correlation with

$$C^A(t) = \langle \psi_A, e^{-tH} \psi_A \rangle, \qquad \psi_A = A\Omega. \tag{7}$$

The spectral theorem gives

$$C^A(t) = \int_{[0,\infty)} e^{-tE}\, d\mu_A(E). \tag{8}$$

If μ_A assigned positive weight to an interval below m_*, its contribution would decay more slowly than e^{-mt} for some $m < m_*$, contradicting (5). Hence every centered vector generated by the designated observables has spectral support in $[m_*, \infty)$. Density excludes an additional physical spectral subspace in $(0, m_*)$. □

Remark 1. This theorem is conditional and belongs to the established family of spectral–clustering principles. Its role here is architectural: it states exactly which rigidity information must be transported through the continuum construction. It does not generate the required estimate.

4 The real problem: generation of the rigidity estimate

The difficult step is now isolated:

> Derive a volume-uniform exponential-decay estimate, in physical units, from local Yang–Mills dynamics along the four-dimensional continuum trajectory.

At fixed strong lattice coupling, cluster expansions, stochastic dynamics, and functional inequalities can yield infinite-volume limits and exponential decay. Recent Poincaré and log-Sobolev approaches give such results for lattice Yang–Mills in specified strong-coupling regimes [3, 4]. These are genuine lattice correlation-decay results. They do not reach the weak bare-coupling limit $\beta \to \infty$ associated with the four-dimensional continuum construction.

The missing inference is therefore not the formal implication

$$\text{finite lattice gap} \longrightarrow \text{continuum gap}. \tag{9}$$

It is the dynamical statement

$$\boxed{\text{asymptotically free ultraviolet theory} \longrightarrow \text{controlled entry into an infrared rigidity regime.}} \tag{10}$$

5 A two-regime proof architecture

The representation ontology suggests separating ultraviolet transport from infrared rigidity generation.

5.1 Regime I: ultraviolet coherence transport

Beginning with a fine Wilson lattice, one seeks to

1. perform gauge-covariant block transformations;
2. integrate out short-distance degrees of freedom;
3. transport a convergence-determining gauge-invariant observable algebra;
4. preserve reflection positivity or prove an adequate recovery theorem;
5. control small-field and large-field sectors separately;
6. retain explicit dependence on lattice spacing and running coupling.

The intended chain is

$$\mathfrak{R}_a \longrightarrow \mathfrak{R}_{ba} \longrightarrow \mathfrak{R}_{b^2a} \longrightarrow \cdots \longrightarrow \mathfrak{R}_{\ell_*}. \tag{11}$$

Balaban's four-dimensional analysis provides rigorous small-field renormalization-group structure, but it does not complete the continuum construction or establish the infrared gap [2].

5.2 Regime II: infrared rigidity generation

At a dynamically generated physical scale ℓ_*, the effective gauge-invariant representation should enter a domain in which one can prove conditional mixing, a block functional inequality, transfer-operator contraction, effective coercivity, or another local positive inequality that implies exponential correlation decay.

A candidate block inequality is

$$\mathrm{Var}_{\mu_*}(F) \le \frac{1}{\lambda_*} \sum_B \mathbb{E}_{\mu_*} \left[\mathrm{Var}(F \mid \mathcal{F}_{B^c}) \right], \qquad \lambda_* > 0, \tag{12}$$

together with sufficiently weak inter-block influence. The desired consequence is

$$|\mathrm{Cov}_{\mu_*}(A, B)| \le C_{A,B} e^{-c\, d(A,B)/\ell_*}, \tag{13}$$

giving the physical rate

$$m_* = \frac{c}{\ell_*}. \tag{14}$$

A Poincaré inequality for an auxiliary Langevin dynamics is not itself the quantum Hamiltonian mass gap. It is an analytic engine for deriving physical correlation decay. The rigidity-transfer theorem must then carry that decay to the OS-reconstructed Hamiltonian.

6 The central missing theorem: infrared basin entry

The interpretation identifies a sharper frontier target than the undivided instruction to "prove the mass gap."

Conjectural theorem target 1 (Yang–Mills Infrared Basin Entry)**.** *The gauge-covariant renormalization flow of four-dimensional pure Yang–Mills theory, starting on the asymptotically free*

continuum trajectory, reaches at a finite physical scale ℓ_* a gauge-invariant effective representation satisfying a verifiable local rigidity criterion:

$$\mathfrak{R}_{a,g(a)} \xrightarrow[a\to 0]{\text{controlled RG}} \mathfrak{R}_{\ell_*} \in \mathfrak{B}_{\text{rigid}}. \tag{15}$$

Here $\mathfrak{B}_{\text{rigid}}$ is defined by explicit local inequalities and does not assume a mass gap or exponential clustering.

A proof would have to show:

1. ℓ_* remains finite and nonzero in physical units;
2. the effective representation retains the correct Yang–Mills observable content;
3. reflection positivity or a sufficient substitute survives;
4. the rigidity criterion does not contain clustering among its premises;
5. large-field and topological sectors are controlled;
6. the constants are uniform in the original volume;
7. pulling the bound back through the RG chain does not make it vanish.

Perturbative renormalization suggests a scale of the schematic form

$$\ell_* \sim \mu^{-1} \exp\!\left(\frac{1}{2b_0 g^2(\mu)}\right), \tag{16}$$

but the running-coupling formula does not prove that the full theory enters a massive infrared basin at that scale. The basin-entry theorem is where dimensional transmutation must become a nonperturbative mathematical statement.

7 The complete representation atlas

The proposed proof chain is

$$\begin{aligned}\mathfrak{R}_{\text{Wilson}} &\xrightarrow{\text{gauge-invariant RG}} \mathfrak{R}_{\text{block}} \xrightarrow{\text{basin entry}} \mathfrak{R}_{\text{rigid}} \\ &\xrightarrow{\text{uniform clustering}} \mathfrak{R}_{\text{Euclidean}} \xrightarrow{\text{continuum coherence}} \mathfrak{R}_{\text{OS}} \xrightarrow{\text{reconstruction}} \mathfrak{R}_{\text{Hamiltonian}}.\end{aligned} \tag{17}$$

Each arrow requires an independent certificate.

The ontology prevents a property established in one chart from being silently treated as though it had already survived every subsequent transformation.

8 The role of finite Wilson/OS constructions

A finite Wilson/Osterwalder–Schrader $SU(2)$ construction can serve as a calibrated representation package:

$$\mathfrak{R}_{a,L}^{SU(2)} \longrightarrow \mathcal{H}_{a,L} \longrightarrow H_{a,L}. \tag{18}$$

The Clay-relevant task is to expose the complete dependence

$$\Delta_{a,L} = \Delta\left(a, L, \beta, extgeometry, extobservablesector\right) \tag{19}$$

Representation change	**Required certificate**
Wilson → block variables	Preservation of gauge-invariant observable content
Block variables → rigid regime	Noncircular local coercivity or mixing theorem
Rigid regime → clustering	Quantitative exponential decay in physical units
Regulated theory → continuum	Tightness, renormalization, nontriviality, and symmetry recovery
Euclidean limit → OS theory	Reflection positivity and the remaining OS axioms
OS theory → Hamiltonian gap	Spectral completeness and rigidity transfer

Table 1: The certificates required along the proposed proof chain.

and determine whether

$$\liminf_{\substack{a\to 0\\ L\to\infty}} \frac{\Delta_{a,L}^{\text{lattice}}}{a} > 0. \tag{20}$$

The finite construction becomes relevant to the continuum problem if it provides at least one of the following:

- a physical-unit lower bound uniform in volume;
- a block-local inequality stable under renormalization;
- an exact comparison between transfer spectra and gauge-invariant correlation decay;
- a geometric coercivity estimate whose constants survive continuum scaling;
- control of large-field corrections to a heat-kernel or small-field representation.

Its role is therefore not merely to display a finite eigenvalue separation, but to serve as a laboratory for discovering a transportable rigidity certificate.

9 What may change and what must survive

The proof may change gauge coordinates, lattice variables, gauge fixing, regulators, block variables, local parametrizations, particle interpretations, and apparent microscopic ontology. It must preserve or recover

- gauge-invariant observable content;
- reflection positivity and physical positivity;
- locality or causal reconstructibility;
- four-dimensional continuum covariance;
- asymptotic-freedom scaling;
- nontriviality of the limiting theory;
- completeness of the physical sectors;
- a positive rigidity rate in physical units.

This permits a search for nonclassical variables without changing the target theory, provided every representational bridge is certified.

10 The irreducible core and theorem sequence

The ontology does not dissolve the Clay problem. It exposes its irreducible core:

Prove that four-dimensional non-Abelian RG flow generates a finite physical scale at which gauge-invariant relat

(21)

The research programme should pursue three theorem targets in order:

1. **Rigidity Transfer Theorem:** uniform decay on a complete regulated observable class implies a continuum Hamiltonian gap.

2. **RG-to-Rigidity Theorem:** entry into an explicitly defined functional-inequality or mixing domain implies uniform physical relational rigidity.

3. **Yang–Mills Basin Entry Theorem:** the four-dimensional pure Yang–Mills RG trajectory enters that domain at a finite dynamically generated scale.

The first target is tractable as a precise transport theorem. The second can be proved and stress-tested in restricted models. The third is the genuine frontier theorem. Its validity depends on formulating $\mathfrak{B}_{\text{rigid}}$ through independently verifiable local inequalities rather than by hiding the desired mass gap in its definition.

References

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